DNS walkthrough

The Domain Name System (DNS) is the internet's directory. People use readable names such as `azizulabedinazmi.netlify.app`, while computers connect to servers using IP addresses. DNS translates the name into the address needed to load the website.

When someone enters the site address in a browser, the browser first checks its own cache. If it does not already know the answer, the device asks a DNS resolver, usually provided by the internet service provider or a public DNS service. The resolver looks for the answer, using its cache when possible. Otherwise, it follows the DNS hierarchy: a root server points it to the appropriate top-level-domain nameserver, and that nameserver points it to the domain's authoritative nameserver. The authoritative nameserver stores the actual DNS records and returns the result to the resolver. The resolver sends that result back to the device, which can then connect to the hosting provider. The browser uses HTTPS to request the page, and the host returns the site's files.

A DNS record is an instruction about a domain. An `A` record maps a name directly to an IPv4 address. A `CNAME` record maps one hostname to another hostname instead. For example, a custom `www` hostname can use a CNAME that points to a hosting provider's assigned hostname. The resolver follows that alias and then looks up the final address. CNAMEs are useful when the host manages changing server addresses, because the site owner does not have to update an IP address manually.

For this assignment, Netlify provides the public `netlify.app` hostname and automatically serves the site over HTTPS. A custom domain is optional. If I connect one later, I would add the DNS records requested by Netlify at my domain registrar, wait for the records to propagate through resolver caches, and then verify that the custom address loads with a valid HTTPS certificate.